13.4.2 Methods of preparation of cyanides

1) From alkyl halides

When alkyl halides are treated in the solution NaCN (or) KCN , alkyl cyanides are obtained. In this reaction a new carbon – carbon bond is formed.

Example

$$KCN + CH_3CH_2 - Br \longrightarrow CH_3CH_2 - CN + KBr$$

Ethyl bromide — Propanenitrile

Aryl cyanide cannot be prepared in this method because of their less reactivity towards nucleophilic substitution. Aryl cyanides are prepared using Sandmeyers reactions.

2. By dehydration of primary amides and aldoximes with P_2O_5

$$CH_3 - CONH_2 \xrightarrow[-H_2O]{P_2O_5} CH_3 - CN$$

Acetamide — Ethanenitrile

$$CH_3 - CH{=}NOH \xrightarrow[-H_2O]{P_2O_5} CH_3 - CN$$

Acetaldoximes

3. By dehydration of ammonium carboxylates with P_2O_5

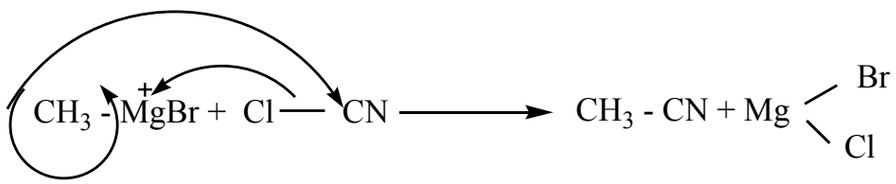
$$CH_3 - COONH_4 \xrightarrow[\Delta]{P_2O_5} CH_3 - CN + 2H_2O$$

Ammonium acetate — Ethanenitrile

This method suitable for large scale preparation of alkyl cyanides.

4. From Grignard reagent

Methyl magnesium bromide on treatment with cyanogen chloride (Cl - CN) forms ethanenitrile.



methyl magnesium bromide — ethanenitrile

13.4.3 Properties Of Cyanides

Physical Properties

The lower members (up to C_{14}) are colourless liquids with a strong characteristic sweet smell. The higher members are crystalline solids, They are moderately soluble in water but freely souble in organic solvents. They are poisonous.

They have higher boiling points than analogous acetylenes due to their high dipole moments.

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